Appendix: Theoretical Analysis of the Augmented Synthetic Historical Control Estimator (ASHC)

Setup and Assumptions

For the purposes of self-containment, let $\mathcal{T} = \{1, 2, \ldots, T\}$ index time at a monthly frequency. Define the pre-treatment period as $\mathcal{T}_1 := \{t \in \mathcal{T} : t \leq T_0\}$ and the post-treatment period as $\mathcal{T}_2 := \{t \in \mathcal{T} : t > T_0\}$. The number of post-treatment periods is $n = T - T_0$. Let $m \in \mathbb{N}$ denote the evaluation window length. Define the evaluation period as the final m months of the pre-treatment period $\mathcal{T}_{\text{eval}} := \{T_0 - m + 1, \ldots, T_0\} \subset \mathcal{T}_1$.

Let $\mathbf{y} = (y_1, y_2, \ldots, y_T)^\top \in \mathbb{R}^T$ denote the observed outcome vector for the treated unit. Define the pre-treatment outcome vector $\mathbf{y}_{\text{pre}} := \mathbf{y}_{\mathcal{T}_1} \in \mathbb{R}^{T_0}$, the evaluation subvector $\mathbf{y}_{\text{eval}} := \mathbf{y}_{\mathcal{T}_{\text{eval}}} \in \mathbb{R}^m$, and the post-treatment outcome vector $\mathbf{y}_{\text{post}} := \mathbf{y}_{\mathcal{T}_2} \in \mathbb{R}^n$.

The evaluation subvector $\mathbf{y}_{\text{eval}}$ is the target pattern to be reconstructed using convex combinations of earlier segments from the pre-treatment period.

SHC by Chen et al. [2024] constructs

$$N = T_0 - m - n + 1$$

overlapping temporal segments of length m. Each segment $\widetilde{\mathbf{y}}_{[i,i+m-1]}$ is a contiguous subvector of the smoothed pre-treatment series. These comprise the donor matrix

$$\widetilde{\mathbf{Y}}_{\text{pre}} := \begin{bmatrix} \widetilde{\mathbf{y}}_{[1,m]} & \widetilde{\mathbf{y}}_{[2,m+1]} & \cdots & \widetilde{\mathbf{y}}_{[N,N+m-1]} \end{bmatrix} \in \mathbb{R}^{m \times N}.$$

Example 0.1. *Suppose we have $T_0 = 60$ months of pre-treatment data and we wish to construct donor segments of length $m = 12$ months to predict a post-treatment horizon of $n = 6$ months. The number of eligible segments is:*

$$N = 60 - 12 - 6 + 1 = 43.$$

Each donor segment is a vector of 12 consecutive smoothed pre-treatment outcomes.

In the third step, we solve the following program to obtain the optimal weights:

$$\begin{aligned} \mathbf{w}^* = \underset{\mathbf{w} \in \mathbb{R}^N}{\operatorname{argmin}} \quad & \left\| \widetilde{\mathbf{y}}_{\text{eval}} - \widetilde{\mathbf{Y}}_{\text{pre}} \mathbf{w} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2 \\ \text{subject to} \quad & \mathbf{w} \geq \mathbf{0}, \quad \mathbf{1}^\top \mathbf{w} = 1. \end{aligned}$$

Let $\mathbf{w}^{\text{SHC}}$ denote the SHC solution, and $\mathbf{w}^{\text{ASHC}}$ the ASHC solution:

$$\mathbf{w}^{\text{ASHC}} = \arg\min_{\mathbf{w} \in \mathbb{R}^N} \left\{ \frac{1}{2\lambda} \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}\|^2 + \|\mathbf{w} - \mathbf{w}^{\text{SHC}}\|^2 + \varsigma \|\mathbf{C}_0^\top \mathbf{w}\|_2^2 \right\} \quad \text{s.t.} \ \sum_{i=1}^N w_i = 1,$$

where $\lambda > 0$ is the regularization parameter, $\varsigma > 0$ is the penalty parameter for the low-variance directions, and $\mathbf{C}_0 \in \mathbb{R}^{N \times k}$ contains the eigenvectors of the Gram matrix $\mathbf{G} = \widetilde{\mathbf{Y}}^\top \widetilde{\mathbf{Y}}$ corresponding to eigenvalues less than a threshold $\tau > 0$.

Assumptions

Assumption 0.1 (Linearity with noise)**.** *There exists* $\mathbf{w}^* \in \mathbb{R}^N$ *such that*

$$\widetilde{\mathbf{y}} = \widetilde{\mathbf{Y}}\mathbf{w}^* + \boldsymbol{\eta},$$

where $\boldsymbol{\eta} \in \mathbb{R}^m$ *is a mean-zero noise vector.*

Assumption 0.2 (Bounded noise)**.** $\|\boldsymbol{\eta}\| \le \delta$ *for some* $\delta > 0$*.*

Assumption 0.3 (Smoothness of latent trend)**.** *Let* $\ell(t)$ *denote the latent smooth component of the treated potential outcome. Suppose* $\ell(t)$ *is* $H+1$ *times differentiable with* $|\ell^{(H+1)}(t)| < \epsilon$ *for all* t *and some* $H \ge 3$*.*

Assumption 0.4 (Operator norm bound)**.** $\|\widetilde{\mathbf{Y}}\| \le \gamma$*.*

Assumption 0.5 (Simplex feasibility)**.** $\mathbf{w}^{SHC} \in \Delta^{N-1} := \{\mathbf{w} \in \mathbb{R}^N : \sum w_i = 1,\ w_i \ge 0\}$*.*

Lemma 0.1 (Deviation from SHC)**.** *Let* $\Delta := \mathbf{w}^{ASHC} - \mathbf{w}^{SHC}$*. Then*

$$\|\Delta\| \ \le \ \frac{\epsilon_{SHC}}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}},$$

where $\epsilon_{SHC} := \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{SHC}\|$ *and* $\|\mathbf{C}_0\|^2$ *is the operator norm of* $\mathbf{C}_0^\top \mathbf{C}_0$*.*

Proof. To bound in sample deviation, we can begin from the ASHC optimization problem:

$$J(\mathbf{w}) = \frac{1}{2\lambda}\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}\|^2 + \|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\|^2 + \varsigma\|\mathbf{C}_0^\top \mathbf{w}\|_2^2 \quad \text{s.t. } \mathbf{1}^\top \mathbf{w} = 1.$$

The three components have clear roles: the first encourages close fit to the evaluation period outcomes, the second penalizes deviation from the SHC weights, and the third stabilizes the solution in low-variance directions.

If we minimized $J(\mathbf{w})$ without the affine constraint $\mathbf{1}^\top \mathbf{w} = 1$, the resulting $\mathbf{w}$ might not be feasible. To enforce this constraint, we introduce the Lagrangian

$$\mathcal{L}(\mathbf{w}, \mu) = J(\mathbf{w}) + \mu(\mathbf{1}^\top \mathbf{w} - 1),$$

where $\mu \in \mathbb{R}$ is a Lagrange multiplier. This term acts as a corrective force, ensuring feasibility without changing the curvature of $J(\mathbf{w})$ that governs the size of Δ. We now may compute the gradient of $\mathcal{L}$.

First, for the fit term, define $r(\mathbf{w}) = \widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}$. Using the *chain rule for vector functions*,

$$\nabla_{\mathbf{w}} \frac{1}{2\lambda}\|r(\mathbf{w})\|^2 = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top(\widetilde{\mathbf{Y}}\mathbf{w} - \widetilde{\mathbf{y}}).$$

Second, for the deviation penalty, recall the rule for the gradient of a quadratic form, $\nabla_{\mathbf{w}}(\mathbf{w}^\top A\mathbf{w}) = (A + A^\top)\mathbf{w}$. With $A = I$, we obtain

$$\nabla_{\mathbf{w}}\|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\|^2 = 2(\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}).$$

Third, for the low-variance penalty,

$$\varsigma\|\mathbf{C}_0^\top \mathbf{w}\|^2 = \varsigma \mathbf{w}^\top(\mathbf{C}_0\mathbf{C}_0^\top)\mathbf{w}.$$

Since $\mathbf{C}_0\mathbf{C}_0^\top$ is symmetric, applying the quadratic form rule yields

$$\nabla_{\mathbf{w}}\varsigma\|\mathbf{C}_0^\top \mathbf{w}\|^2 = 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top \mathbf{w}.$$

Finally, differentiating the affine constraint term $\mu(\mathbf{1}^\top \mathbf{w} - 1)$, linearity of differentiation gives

$$\nabla_{\mathbf{w}}[\mu(\mathbf{1}^\top \mathbf{w} - 1)] = \mu\mathbf{1}.$$

Collecting terms, the stationarity condition $\nabla_{\mathbf{w}}\mathcal{L}(\mathbf{w},\mu)=0$ is

$$\left(\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}}+2\mathbf{I}+2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\right)\mathbf{w}=\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{y}}+2\mathbf{w}^{\mathrm{SHC}}-\mu\mathbf{1}.$$

Let $M:=\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}}+2\mathbf{I}+2\varsigma\mathbf{C}_0\mathbf{C}_0^\top$. Subtracting the equation at $\mathbf{w}^{\mathrm{SHC}}$, we isolate the deviation $\Delta=\mathbf{w}^{\mathrm{ASHC}}-\mathbf{w}^{\mathrm{SHC}}$:

$$M\Delta=\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top(\widetilde{\mathbf{y}}-\widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}})-\mu\mathbf{1}.$$

The affine constraint ensures $\mathbf{1}^\top\Delta=0$, so Δ lies in the hyperplane orthogonal to $\mathbf{1}$. This means the term $-\mu\mathbf{1}$ does not affect the component of Δ in this subspace: μ enforces feasibility but does not change the bound on $\|\Delta\|$.

Let $r:=\widetilde{\mathbf{y}}-\widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}}$ with $\|r\|=\epsilon_{\mathrm{SHC}}$ be the SHC residual. Then

$$\|M\Delta\|\le\frac{1}{\lambda}\|\widetilde{\mathbf{Y}}^\top r\|\le\frac{\gamma}{\lambda}\epsilon_{\mathrm{SHC}},$$

by Assumption (A4). Since the minimum eigenvalue of M is at least $2+2\varsigma\|\mathbf{C}_0\|^2$, we invert M to obtain

$$\|\Delta\|\le\frac{\gamma}{2\lambda+2\varsigma\|\mathbf{C}_0\|^2}\,\epsilon_{\mathrm{SHC}}.$$

This establishes the claimed bound. □

Remark 0.1. *The key role of the affine constraint $\mathbf{1}^\top\mathbf{w}=1$ is to introduce a Lagrange multiplier μ, which shifts the optimality condition by a vector in the span of $\mathbf{1}$. Because Δ must satisfy $\mathbf{1}^\top\Delta=0$, this shift does not affect the conditioning of M and thus does not alter the deviation bound. The effective regularization is governed entirely by the spectrum of M.*

Lemma 0.2. *Under (A4), the error from deviating off SHC weights satisfies*

$$\|\widetilde{\mathbf{Y}}\Delta\|\le\gamma\|\Delta\|.$$

Remark 0.2. *Lemma 0.2 ensures that deviations of the ASHC weights from the SHC weights, measured in ℓ_2 norm, translate into controlled deviations in the predicted outcomes. The operator norm bound (A4) prevents $\widetilde{\mathbf{Y}}$ from amplifying small changes in the weights disproportionately.*

Lemma 0.3. *Under (A3), for horizon $k>0$,*

$$|\ell(T_0+k)-\ell^{SHC}(T_0+k)|\le b_\epsilon(H,k):=\frac{2\epsilon|k|^{H+1}}{(H+1)!}.$$

Proof. Let $\ell(t)$ denote the latent smooth component of the treated outcome. By (A3), $\ell(t)$ is $(H+1)$ times differentiable with $|\ell^{(H+1)}(t)|\le\epsilon$.

Step 1. Taylor expansion around T_0. By Taylor's theorem,

$$\ell(T_0+k)=\sum_{h=0}^{H}\frac{\ell^{(h)}(T_0)}{h!}\,k^h+R_{H+1}(k),$$

with remainder

$$R_{H+1}(k)=\frac{\ell^{(H+1)}(\xi)}{(H+1)!}\,k^{H+1},\quad\text{for some }\xi\in[T_0,T_0+k].$$

Step 2. Bound the remainder.

$$|R_{H+1}(k)| \le \frac{\epsilon}{(H+1)!}\,|k|^{H+1}.$$

Step 3. SHC approximation of lower-order terms. By construction, SHC matches the first H terms of this expansion up to error of the same order as the remainder.

Step 4. Combine bounds.

$$|\ell(T_0+k) - \ell^{\mathrm{SHC}}(T_0+k)| \le \frac{2\epsilon}{(H+1)!}\,|k|^{H+1}.$$

□

Remark 0.3. *Lemma 0.3 formalizes the SHC bias as a Taylor remainder. The SHC approximation matches derivatives of $\ell(t)$ at T_0 up to order H, leaving bias decaying like $|k|^{H+1}$. Thus, for smooth latent trends and moderate horizons, SHC bias is negligible.*

Main Result

Proposition 0.1. *Under assumptions (A1)–(A5), the ASHC prediction error is bounded by*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{ASHC}\| \le \epsilon_{SHC}\left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}}\right) + b_\epsilon(H,k) + \delta.$$

Proof. We want to bound the prediction error of the ASHC estimator,

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\|.$$

Step 1: Decompose the error using the triangle inequality. Let $\mathbf{w}^*$ denote the ideal weights from the SHC estimator without the ASHC penalty. Define the difference $\Delta := \mathbf{w}^{\mathrm{ASHC}} - \mathbf{w}^*$. Then,

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| = \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^* - \widetilde{\mathbf{Y}}\Delta\| \le \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^*\| + \|\widetilde{\mathbf{Y}}\Delta\|.$$

The first term, $\epsilon_{\mathrm{SHC}} := \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^*\|$, is the SHC prediction error without augmentation.

Step 2: Control the second term using operator norm. By Lemma 0.2, the matrix $\widetilde{\mathbf{Y}}$ satisfies

$$\|\widetilde{\mathbf{Y}}\Delta\| \le \gamma\|\Delta\|,$$

where γ is an operator norm bound capturing how the weights perturbations translate into prediction changes.

Step 3: Bound the weight difference $\|\Delta\|$. From Lemma 1 (stated earlier), the ASHC weight vector $\mathbf{w}^{\mathrm{ASHC}}$ stays close to the SHC weights in Euclidean norm:

$$\|\Delta\| = \|\mathbf{w}^{\mathrm{ASHC}} - \mathbf{w}^*\| \le \frac{\epsilon_{\mathrm{SHC}}}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}},$$

where the denominator reflects the strength of the ASHC penalty, ensuring stability.

Step 4: Incorporate smoothness bias and noise. Finally, the total prediction error also includes the smoothness bias $b_\epsilon(H,k)$ due to approximating the latent function by a truncated Taylor expansion of order H, and a noise term bounded by δ (from assumption (A2)).

Putting all these together, we get

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \le \epsilon_{\mathrm{SHC}} + \gamma\|\Delta\| + b_\epsilon(H,k) + \delta \le \epsilon_{\mathrm{SHC}}\left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}}\right) + b_\epsilon(H,k) + \delta,$$

which completes the proof. □

Discussion

Proposition~0.1 decomposes ASHC error into: 1. SHC fit error (ϵ_{SHC}), 2. amplification from ASHC deviation, 3. smoothness bias $b_\epsilon(H,k)$, 4. noise δ.

The penalty term $\varsigma\|\mathbf{C}_0^\top \mathbf{w}\|_2^2$ reduces deviation but may bias if $\mathbf{w}^*$ has components in $\mathbf{C}_0$ directions. The trade-off between λ and ς governs stability vs. fidelity.

Asymptotic Results

Theorem 0.1. *Under (A1)–(A5), as $m \to \infty$ and $H \to \infty$, for fixed $k > 0$,*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{ASHC}\| \longrightarrow \delta.$$

If $\boldsymbol{\eta}$ consists of independent mean-zero sub-Gaussian entries, then

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{ASHC}\| \xrightarrow{p} 0.$$

Proof. Starting from Proposition 0.1, we have

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{\text{ASHC}}\| \le \epsilon_{\text{SHC}} \left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}}\right) + b_\epsilon(H,k) + \delta.$$

Step 1. Behavior of ϵ_{SHC}. By (A3), $\ell(t)$ is $(H+1)$-times differentiable. Gaussian kernel smoothing in SHC ensures

$$\epsilon_{\text{SHC}} \to 0 \quad \text{as } m \to \infty,\ H \to \infty.$$

Chen et al. [2024]

Step 2. Behavior of ASHC penalty term. Since $\epsilon_{\text{SHC}} \to 0$, the inflation factor

$$\frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}}\,\epsilon_{\text{SHC}}$$

vanishes asymptotically.

Step 3. Behavior of smoothness bias $b_\epsilon(H,k)$. From Taylor's theorem,

$$\ell(T_0 + k) = \sum_{h=0}^{H} \frac{\ell^{(h)}(T_0)}{h!} k^h + R_{H+1}(k),$$

with

$$|R_{H+1}(k)| \le \frac{\epsilon}{(H+1)!}|k|^{H+1}.$$

Thus

$$b_\epsilon(H,k) \le \frac{2\epsilon}{(H+1)!}|k|^{H+1} \to 0 \quad \text{as } H \to \infty.$$

Step 4. Behavior of noise term. By (A2), $\|\boldsymbol{\eta}\| \le \delta$. If $\boldsymbol{\eta}$ is i.i.d. sub-Gaussian with variance proxy σ^2, then

$$\frac{\|\boldsymbol{\eta}\|}{m} \xrightarrow{p} 0.$$

Step 5. Combine bounds. Therefore,

$$\limsup_{m\to\infty} \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{\text{ASHC}}\| \le \delta.$$

With sub-Gaussian noise, the limit is 0 in probability. □

Corollary 0.1. *Under (A1)–(A5), the ASHC estimator is asymptotically unbiased up to δ. If $\boldsymbol{\eta}$ is i.i.d. mean-zero sub-Gaussian with variance proxy σ^2 and bandwidth $h_m \asymp m^{-1/(2H+3)}$, then as $m \to \infty$,*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{ASHC}\| = O_p\Big(m^{-H/(2H+3)}\Big).$$

For large H, this approaches the parametric rate $O_p(m^{-1/2})$.

Proof. **Part 1: Consistency.** From Theorem 0.1, the error converges to δ, or 0 in probability under sub-Gaussian noise.

Part 2: Convergence Rate. From Proposition 0.1, we have

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{\text{ASHC}}\| \le \epsilon_{\text{SHC}} \left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}}\right) + b_\epsilon(H,k) + \delta.$$

Step 1. Bias term. From Lemma 0.3,

$$b_\epsilon(H,k) = O(h_m^H).$$

Step 2. Stochastic error term. With sub-Gaussian noise, the smoothed variance is

$$O_p\big((mh_m)^{-1/2}\big).$$

Step 3. SHC fit error. Combining,

$$\epsilon_{\text{SHC}} = O_p\big(h_m^H + (mh_m)^{-1/2}\big).$$

Step 4. Optimal bandwidth. Equating orders $h_m^H \asymp (mh_m)^{-1/2}$ yields

$$h_m \asymp m^{-1/(2H+3)}.$$

Step 5. Convergence rate. Substituting,

$$\epsilon_{\text{SHC}} = O_p\big(m^{-H/(2H+3)}\big).$$

Thus the total error is

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{\text{ASHC}}\| = O_p\big(m^{-H/(2H+3)}\big).$$

Step 6. Near-parametric rate. As $H \to \infty$,

$$\frac{H}{2H+3} \to \tfrac{1}{2}.$$

So the rate approaches $O_p(m^{-1/2})$. □

Remark 0.4. *The deterministic noise bound δ in (A2) corresponds to $\|\boldsymbol{\eta}\| \le \delta$. Under sub-Gaussian noise, $\delta = O(\sigma\sqrt{m})$, while kernel smoothing reduces effective noise to $O_p((mh_m)^{-1/2})$. This explains the appearance of the nonparametric rate.*

References

Yi-Ting Chen, Jui-Chung Yang, and Tzu-Ting Yang. Synthetic historical control for policy evaluation. Available at SSRN: https://ssrn.com/abstract=4995085, September 2024. URL http://dx.doi.org/10.2139/ssrn.4995085.